

All-Period Integral Monodromy and Algebraic-Unit Multipliers for the Area-Preserving Hénon Map

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Abstract

Pressure normalization gives a prime-orbit counting law for a certified Hénon suspension, leaving the arithmetic type of its orbit labels as the next gate. We prove an all-period algebraic theorem for the underlying autonomous map $H_6(q, p) = (1 - 6q^2 - p, q)$. After the canonical scaling $x_i = 6q_i$, the period- n fixed algebra is defined by monic cyclic quadratics and is finite free over $\mathbb{Z}$ of rank 2^n . Every chronological derivative step lies in SL_2 over this integral algebra. Therefore every periodic monodromy trace is an algebraic integer and every multiplier is an algebraic unit. A separate exact implementation verifies the Gröbner and trace ledgers through period 10 and recovers the inherited degree-four fixed-point multiplier polynomial. The theorem supplies genuine all-period arithmetic structure, but it also rules out treating a raw multiplier or its field norm as a rational prime. It does not classify the nonalgebraic pressure power $|\Lambda|^{h_*}$.

1 Integral cyclic coordinates

The Hénon recurrence is

$$q_{i+1} = 1 - 6q_i^2 - q_{i-1}.$$

Set $x_i = 6q_i$. Then

$$x_{i+1} = 6 - x_i^2 - x_{i-1}. \tag{1}$$

For period n , indices are read cyclically. The neighbor multiset must be retained at low period, giving

$$n = 1 : \quad x_0^2 + 2x_0 - 6 = 0,$$

and

$$n = 2 : \quad x_0^2 + 2x_1 - 6 = x_1^2 + 2x_0 - 6 = 0.$$

2 Finite-free fixed algebra

Let

$$\mathcal{A}_n = \mathbb{Z}[x_0, \dots, x_{n-1}]/(f_0, \dots, f_{n-1}), \quad f_i = x_i^2 + x_{i-1} + x_{i+1} - 6,$$

with the exceptional neighbor multiplicities above.

Theorem 2.1 (Integral all-period fixed algebra). *For every $n \geq 1$, $\mathcal{A}_n$ is finite free over $\mathbb{Z}$ of rank 2^n , with basis*

$$x_0^{\epsilon_0} \cdots x_{n-1}^{\epsilon_{n-1}}, \quad \epsilon_i \in \{0, 1\}.$$

In particular every coordinate value at every geometric periodic point is an algebraic integer.

Proof. Under any degree-compatible monomial order, f_i is monic with leading monomial x_i^2 . The leading monomials are pairwise coprime, so Buchberger's product criterion makes the displayed equations a monic Gröbner basis over $\mathbb{Z}$. The standard monomials are exactly the square-free monomials, which proves finite freeness and the rank. Specializing a finite integral $\mathbb{Z}$ -algebra at a geometric point sends each x_i to an algebraic integer. $\square$

This strengthens the earlier finite-flat statement over $\mathbb{Z}[A, A^{-1}]$ at the integer specialization: the scaled $A = 6$ fibre needs no denominator localization.

3 Chronological monodromy

In either the original (q_i, q_{i-1}) coordinates or the linearly scaled coordinates, the derivative step is conjugate to

$$J_i = \begin{pmatrix} -2x_i & -1 \\ 1 & 0 \end{pmatrix}, \quad \det J_i = 1. \quad (2)$$

For a period- n point, let

$$M_\gamma = J_{n-1} \cdots J_0, \quad t_\gamma = \operatorname{tr} M_\gamma.$$

Later times act on the left.

Theorem 3.1 (Algebraic-unit multiplier theorem). *For every geometric periodic point of H_6 at every period, t_γ is an algebraic integer and both eigenvalues of M_γ are algebraic units. Their characteristic polynomial is*

$$X^2 - t_\gamma X + 1.$$

Proof. Equation (2) and Theorem 2.1 show that every entry of M_γ , hence its trace, is an algebraic integer. An eigenvalue λ is integral over the ring of algebraic integers by the displayed monic polynomial, hence integral over $\mathbb{Z}$. Since $\det M_\gamma = 1$, the other eigenvalue is λ^{-1} and is also an algebraic integer. Thus λ is a unit. $\square$

The theorem includes nonreal and multiple geometric points scheme by scheme; reducedness is not needed for the integrality statement. On the certified real hyperbolic survivor, the unstable multiplier is the root of modulus greater than one.

4 Exact sentinels

The code builds the Gröbner basis and chronological trace independently for $1 \leq n \leq 10$. The first reduced traces are

$$-2x_0, \quad 4x_0x_1 - 2, \quad -8x_0x_1x_2 + 2(x_0 + x_1 + x_2).$$

All coefficients are integers and all monomials lie in the square-free basis. At period one, $x_0 = -1 \pm \sqrt{7}$ and the two trace-conjugate characteristic polynomials multiply to

$$X^4 - 4X^3 - 22X^2 - 4X + 1,$$

recovering the exact non-lattice witness from the instability-roof project.

5 Arithmetic consequences and boundary

If an algebraic unit lies in $\mathbb{Q}$, it is ± 1 . Thus no hyperbolic raw multiplier can itself be a rational prime. Its field norm is also ± 1 . Trace and $\det(I - M)$ are algebraic integers but are not multiplicative under repetition, so they cannot simply replace the Euler label without a new compiler.

The pressure-normalized label $|\Lambda|^{h^*}$ is deliberately outside this corollary. The pressure root is a real dynamical invariant whose algebraic type is unknown; no valid transcendence conclusion is available here.

6 Evaluator verdict

The all-period unit theorem is positive arithmetic structure at A1/A3, but it constructs no prime correspondence or determinant. The strict tuple is

$$(A1_{\text{WEAK}}, A2_{\text{FAIL}}, A3_{\text{PARTIAL}}, A4_{\text{FORMAL}}),$$

with overall ROUTE_A_EXPLORATORY. Route B is not testable.

7 Conclusion

The Hénon instability multipliers are not arbitrary reals: after the correct integral scaling, they form an all-period family of algebraic units. This both strengthens the arithmetic content of the pressure-normalized road and closes its most naive raw-prime interpretation. The next step is to classify which scalar functions of a unit can preserve exact repetition and still produce rational prime labels.

References

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